Shopping Assistant with Memory

1. Aim

To build a shopping assistant in Langflow Desktop 1.7.2 that accepts shopping requests, remembers earlier conversation context within the same session, and uses that memory to give more relevant and practical suggestions.

2. Objectives

- Understand how a memory-enabled shopping assistant differs from a one-time chatbot response.
- Learn how to create a Langflow workflow using Chat Input, Prompt Template, Message History, OpenRouter, and Chat Output.
- Identify the role of session continuity in preserving shopping context across repeated user interactions.
- Observe how the assistant adapts suggestions after the user states preferences such as healthy snacks.
- Understand the role of prompt design in controlling the behavior of the assistant.

3. Problem Statement

Develop a virtual agent that helps a user build a shopping list across multiple sessions. The agent should remember previous items, user preferences, and adapt its suggestions accordingly.

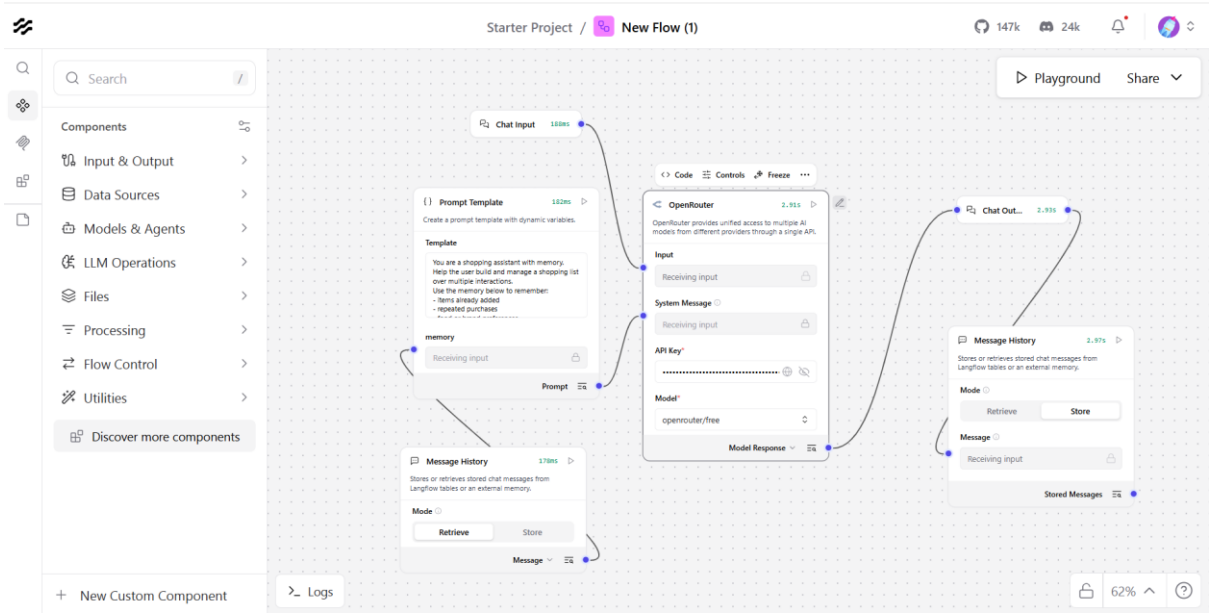
4. Introduction

A practical shopping assistant must do more than answer one isolated question. In real life, people add items gradually, change their mind, state preferences later, and return to the same list more than once. Therefore, a useful assistant should preserve context and refer back to earlier shopping information when the conversation continues.

This experiment demonstrates such behavior using Langflow, a visual workflow environment in which an application is built by connecting components rather than writing a large amount of code. In this lab, the user message enters through Chat Input, prior conversational context is retrieved through Message History, a Prompt Template tells the model how to behave, the OpenRouter component generates the response, and Chat Output displays the result in the Playground.

5. Type Of Workflow/ Agent

This is best described as an LLM-assisted conversational workflow with memory-aware prompting. It is agentic in the sense because it responds to input, uses context, and updates its future behavior. However, it is not a fully autonomous agent that independently purchases products, accesses payment systems, or triggers external tools. It remains a guided conversational assistant.



Default Session

User

add milk, Bread, and apples to my shopping list

AI openrouter/free

I've added milk, bread, and apples to your shopping list.

User

suggest some healthy items and give finalized list

AI openrouter/free

AI: Sure! Based on your preferences, here are some healthy items: fruits, vegetables, whole grains, and lean proteins. Your finalized shopping list might include: milk, bread, apples, fruits, vegetables, and lean proteins.